

Second Assessment – B.E (CSE) – III semester
CS23304 – Java Programming
Common to all Batches

Date: 04.11.25

Max.Marks: 50

Part-A (5X 2 = 10 marks)

1. Explain how to print all the elements in a collection without using an Iterator.
2. Differentiate between collections and streams
3. Write the lifecycle of Servlets
4. Compare and contrast between notify and notifyAll
5. Give a regular expression to generate an email id with the following format: first four letters of first name followed by first 5 letters of second name and date of birth in this format:mm-dd-yyyy @ test.com

Part-B (10 X 3 = 30 marks)

(Answer an THREE)

6. Create a **City** class with name (string), population (double) and stateCapital(Boolean) indicating whether the city is a stateCapital or not and onTheSea (Boolean) – is it seashore city or no and include the necessary methods. Write a test class to print the following using Lambda expressions:
 - a. Print the cities that are on seashore as well as state capital
 - b. Print the cities sorted in alphabetical order and sorted in population
 - c. Print the cities that are stateCapital sorted by population
 - d. Print the cities that are on the sea shore and sort alphabetically
7. An astronomical observatory has **3 telescopes** available for public viewing. Multiple participants arrive at different times and want to view celestial objects through these telescopes. Each telescope can be used by only **ONE participant at a time**. Write a **multithreading java program such that the Participants** should be served synchronously.
8. A custom window shade designer charges a base fee of Rs.50 per shade. In addition, charges are added for certain styles, sizes, and colors as follows:

Styles		Sizes:		Colors	
Regular shades:	Rs.0	25 inches wide:	Rs.0	Natural:	Rs.5
Folding shades:	Rs.10	27 inches wide:	Rs.2	Blue:	Rs.10
Roman shades:	Rs.15	32 inches wide:	Rs.4	Teal:	Rs.15
		40 inches wide:	Rs.6	Red:	Rs.20
				Green:	Rs.25

Create an application that allows the user to select the style, size, color, and number of shades from lists or combo boxes. The total charges should be displayed.

9. Write a servlet program to receive the username and password and display the same.

Part-C (1X 10 = 10 marks)

10. Develop a **Truck Booking Management System** using **HashMap** data structures to efficiently manage bookings.

```
class Truck {
    private String registrationNumber; // e.g., "TN01AB1234"
    private String truckType; // "SMALL", "MEDIUM", "LARGE", "CONTAINER"
    private double capacityTons; // Capacity in tons
    private boolean isAvailable; // Current availability status
    // Constructor and methods
}

class Booking {
    private String bookingId; // e.g., "BOOK001"
    private String truckRegNo;
    private String route; // e.g., "Chennai-Bangalore"
    private double distance; // in kilometers
    private String bookingDate; // Format: "YYYY-MM-DD"
    private double totalCost;
    // Constructor and methods
}

class TruckBookingSystem {
    private HashMap<String, Truck> truckFleet; // RegNo -> Truck
    private HashMap<String, Booking> bookings; // BookingId -> Booking
    private HashMap<String, Double> routeDistances; // Route -> Distance
    // Constructor public TruckBookingSystem() {
    // Initialize all HashMaps
    }
}
```

Implement the following functions

- Add a new truck to the fleet - return true if added successfully, false if truck already exists

```
public boolean addTruck(Truck truck) {
    // Your implementation (2 marks)
}
```

- Add a predefined route with distance

```
public void addRoute(String route, double distance) {
    // Your implementation (2 marks)
}
```

- Book a truck - return Booking object if successful, null if truck not available

```
public Booking bookTruck(String truckRegNo, String route, String bookingDate)
    (2 marks)
```


First Assessment – B.E (CSE) – III semester
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Common to all Branches

Date: 11.09.25

Max.Marks: 50

Part-A (5X 2 = 10 marks)

1. Once an array is created, its size cannot be changed. Does the following code resize the array?

```
int[] num;  
num = new int[10];  
num = new int[20];
```
2. Identify and correct the errors in the following program:

```
public class Test {  
    public static method1(int n, m) {  
        n += m;  
        method2(3.4);  
    }  
    public static int method2(int n) {  
        if (n > 0) return 1;  
        else if (n == 0) return 0;  
        else if (n < 0) return -1;  
    }  
}
```
3. Suppose that s1, s2, s3, and s4 are four strings, given as follows:

```
String s1 = "Welcome to Java";  
String s2 = s1;  
String s3 = new String("Welcome to Java");  
String s4 = "Welcome to Java";
```

What will be the output for the code snippet?

```
s2.equals(s3)  
s1.compareTo(s2)
```
4. Compare and contrast overriding and overloading.
5. List the advantages of using exception handling. $3 \times 10 = 30$

Part-B (~~10~~ X 3 = 30 marks)

Answer any THREE

6. Create a class **Book** that will record the books purchased by a person. It will have the attributes
 - booksPrices—an array of double values that are the prices of all books
 - booksPurchased—the number of books purchased so far
 - maxNumberBP—the maximum number of book purchases that can be recordedand the following methods:
 - Books(max)—a constructor that sets the maximum number of book purchases to max
 - addPrice(p)—adds to an array a price whose value is p
 - getNumberOfBooksPurchased—returns the number of books purchased
 - getTotalBooksPrice—returns the total price of the books purchased
 - getAverageBookPrice()—returns the average price of all the books purchased
 - getCountAbove(p)—returns the number of books that are greater than p in valueCreate a driver class to create object of type **Book** and test all functions.

7. Create an interface `MessageEncoder` that has a single abstract method `String encode(plainText)`, where `plainText` is the message to be encoded and returns the encoded message. Create a class `SubstitutionCipher` that implements the interface `MessageEncoder`. The class variable is `shift(int)` initialized by constructor that takes one parameter called `shift`. Override the method `encode` such that each letter is shifted by the value in `shift`. For example, if `shift` is 3, a will be replaced by d, b will be replaced by e, etc.
 Create a class `ShuffleCipher` that implements the interface `MessageEncoder` with an attribute `shufflenumber` initialized by the one argument constructor. Override the method `encode` so that the message is shuffled `n` times. To perform one shuffle, split the message in half and then take characters from each half alternately. For example, if the message is `abcdefghi`, the halves are `abcde` and `fghi`. The shuffled message is `afbgchdie`.
8. Suppose `songs` is the input String, write a program to prompt the user for a song number and then extract the song number plus the rest of that string's line from a given input string `songs` and print it using class String or String builder methods
 String songs =
 "1. Bow Wow - Fresh Azimiz\n" +
 "2. Weezer - Beverly Hills\n" +
 "3. Dave Matthews Band - Crash Into Me\n" +
 "4. Sheryl Crow - Leaving Las Vegas\n";
9. Implement the following exceptions for the Question no.6:
 - i. The Books(max) constructor throws `IllegalArgumentException` if `max` is initialised with zero and `booksPurchased` is initialised with zero
 - ii. When `booksPrices` is accessed outofbounds throw `ArrayIndexOutOfBoundsException`
 - iii. The driver class throws `InputMismatchException` when the input obtained from user mismatches with the required type
 - iv. `addPrice(p)` throws Custom Exception whenever the value of `p` is negative or zero
 - v. Include finally block wherever neccessary

Part-C (10 marks)

10. Create an abstract class called `Account` that has datamembers: `accno(int)`, `accname(String)` and `balance(double)`, `contactAddress(Address)`. The member methods are: parameterized constructor and `toString` method. abstract methods : `void deposit(double amt)` and `double withdraw(double amt)`
 Derive two classes from class `Account`: `SavingsAccount` and `CheckingsAccount`. The members of `SavingsAccount` are `noofTransactions(int)`, parameterized constructor, overridden methods `toString()`, `deposit(double amt)` that increments balance with `amt`, `withdraw(double amt)` that decrements the balance by `amt` and `noofTransactions` is incremented by 1 inside `deposit()` and `withdraw()`. The members of `CheckingAccount` are parameterized constructor, overridden methods `deposit(double amt)` that increments balance with `amt`, `withdraw(double amt)` that decrements the balance by `amt` only if the balance is above 1000 after decrementation and `toString()`.
 Define `TestAccount` that instantiates objects of `SavingsAccount` and `CheckingsAccount`. The methods of the derived class are tested. Check the `SavingsAccount` object is an instance of class `Account`.